Aman Kumar Sah

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Professional Summary

Full Stack AI Engineer specializing in MERN and Python machine learning pipelines. Experienced in building scalable web applications and deploying AI models to production.

Experience

Junior Research Fellow – Full Stack AI Engineer (MeitY Funded) October 2025 – Present
National Institute of Technology, Raipur *Raipur, C.G.*

- Implemented adversarial debiasing on **Vision Transformer architectures (EVA-02)** using Gradient Reversal Layers to systematically mitigate skin tone classification bias in automated dermatological screening systems.
- Evaluated models against an Indian dermatological dataset, surpassing previous Dermacon IN baselines by improving validation accuracy from **70.16% to 73.68%** and achieving a Disparate Impact ratio of **0.947**.
- Engineered a **genetic algorithm-driven** optimization framework (Python, DEAP) to dynamically adjust thresholds, establishing new benchmarks in algorithmic fairness.
- Architected a **dual-factor biometric** and geofenced attendance monitoring system utilizing FastAPI, React, and DeepFace for real-time tracking and anti-spoofing.

Project Fellow – AI Engineer (NITI Aayog Funded) July 2025 – October 2025
National Institute of Technology, Raipur *Raipur, C.G.*

- Developed the core deep learning classifier model for Smart AgroGuard to automate **AI-based paddy disease detection**.
- Built optimized pipelines for image preprocessing and model evaluation, integrating environmental context temperature fusion to improve overall classifier accuracy from **96.2% to 98.6%**.
- Designed **RESTful APIs** using Python to seamlessly serve the agricultural machine learning predictions to the user-facing mobile application.
- Collaborated with app developers on backend architecture for secure farmer data storage and system authentication.

MERN Stack Developer June 2024 – October 2024
Fixing Dots *Raipur, C.G.*

- Developed the Kearth Solutions company website and a comprehensive admin panel using the **MERN stack** to efficiently manage digital content and business operations.
- Engineered a heavy machinery rental platform, optimizing the login system architecture and database queries to achieve a **92.5% reduction** in authentication response time (from **400ms to 30ms**).

Projects

SecureHostel: Geofenced & Biometric Attendance System | *React, FastAPI, DeepFace* | GitHub

- Architected a dual-factor authentication system integrating DeepFace and Google MediaPipe, engineering dynamic liveness detection to prevent proxy attendance via static images or video.
- Implemented a native GPS geofencing engine utilizing the Haversine formula, alongside an APScheduler-driven daemon to track real-time attendance and trigger automated SMTP alerts.

Intersectional Fairness via Genetic Algorithms | *Python, DEAP* | GitHub

- Developed an intersectional threshold optimization framework using genetic algorithms to dynamically adjust classification decision boundaries for post-processing algorithmic equity.
- Improved Disparate Impact scores over unmitigated baselines by 129% (0.430 to 0.986) on the Adult dataset and 48% (0.650 to 0.964) on COMPAS, while raising baseline accuracy to 73.0% on the German Credit dataset.

Education

Columbia Institute of Engineering and Technology Raipur, C.G.
Bachelor of Technology in Computer Science and Engineering; Score: 83.6% *Aug. 2020 – May 2024*

Shree Shankara Senior Secondary School Urkura, Raipur, C.G.
Class XII; Score: 71.6% *April 2018 – May 2019*

Technical Skills

Languages: Python, JavaScript, C/C++, SQL, HTML/CSS
Web & Backend: React, FastAPI, MongoDB, REST APIs, Node.js, Express.js
AI/ML: PyTorch, Scikit-learn, OpenCV, DeepFace, Computer Vision, Bias Mitigation
Developer Tools: Git, Linux/Ubuntu, VS Code, Postman

Achievements and Publications

- **Publication (Published):** Pradeep Singh, Bikesh Kumar Singh, Rajendra Kumar Pandey, Aman Kumar Sah, "Beyond the Fairness–Accuracy Tradeoff: Genetic Algorithm-Driven Sample Weighting for Age-Based Fairness," *2026 International Conference on Innovative Trends in Information Technology (ICITIIT)*, IEEE, March 2026.
- **Publication (Accepted):** Pradeep Singh, Aman Kumar Sah, Bikesh Kumar Singh, "Multi-attribute AI Fairness via Genetic Algorithm," *2026 IEEE AI Enabled Smart Solutions for Industry, Governance, Healthcare, and Sustainable Development (AISIGH-SD)*, IEEE, Dec 2026.
- **Achievement:** GATE 2026 and GATE 2025 Qualified
- **Achievement:** Student of the Year 2024, Columbia Institute of Engineering and Technology, Raipur